



SOWMYA Y

Data Engineer



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About Me: Results-driven Data enthusiast with 9 years of experience transforming complex data into actionable business insights, consistently delivering improvements in **business outcomes** through **data-driven** solutions. Proficient in **SQL, Python, Tableau**, and **advanced statistical techniques**, seeking to contribute analytical expertise to support strategic objectives in a dynamic organization.

Profile Summary

- Expert in **SQL** query optimization and **Python (Pandas, NumPy, Matplotlib)** for **data manipulation** and analysis, with proven success in handling datasets of 1M+ records while reducing processing time by 20%.
- Advanced practitioner of **data visualization** tools (**Tableau, Power BI, Looker**), creating intuitive dashboards that increased **stakeholder engagement** and streamlined decision-making processes.
- Skilled in designing and implementing **ETL** pipelines using **AWS, Snowflake, and Informatica**, ensuring seamless **data integration** and transformation while maintaining 99.9% data accuracy.
- Experienced in **statistical analysis** and **machine learning**, applying regression, classification, and time series models to uncover business trends and predict customer behaviour.
- Strong background in process automation using **Python, VBA, and Alteryx**, turning manual workflows into efficient automated solutions that reduced operational overhead.
- Streamlined deployment processes by integrating **Docker with CI/CD** pipelines, enabling automated builds, testing, and deployment of data processing jobs.
- Expertise in **SQL** for **database querying, data extraction, and optimization** of complex queries and Skilled in **Python** for **data cleaning, statistical analysis, and machine learning** implementations.
- Adept at performing **predictive analytics** using statistical tools like **R and SAS** and Strong understanding of **data warehousing** concepts and experience working with platforms like **Snowflake and AWS Redshift**.
- Knowledge of **cloud technologies**, including **AWS, Azure, and GCP**, for managing and analysing big data.
- Proficient in **big data** technologies like **Hadoop, Spark, and Kafka** for processing and analysing large-scale datasets.
- Experience building and integrating **RESTful APIs** to enable seamless data flow between services.
- Experience with **API** integration for extracting data from external sources and ensuring seamless data flow.
- Adept at designing **advanced statistical models** and **hypothesis testing** to support business strategy using tools like **SciPy**.
- Extensive experience in working with **Snowflake for data warehousing**, including building scalable data models, optimizing queries, and implementing secure data sharing across multiple teams.
- Expertise in **advanced Excel** functionalities, including pivot tables, **Power Query, complex formulas, macros, and VBA scripting**, to streamline data analysis and automate reporting processes.

Certifications

- Microsoft Certified Power BI Data Analyst.
- Microsoft Azure Data Engineer Associate DP-203.

- Microsoft Ignite Edition Challenge
- Google Data Analytics Professional.
- Generative AI Fundamentals-Databricks
- BCG Gen AI Job Simulation-Forage
- JPMorgan Chase & Co. - Quantitative Research Job Simulation-Forage
- Data analytics and Visualization Job Simulation-Forage
- Introduction to Data Science-Cisco

Technical Skills

Programming Languages: Python, R (Shiny), SQL, JavaScript, Java, C, VBA, PowerShell

Python Libraries: Pandas, NumPy, Matplotlib, Seaborn, SciPy, Stats models, Report Lab, Beautiful Soup, PySpark, XGBoost, TensorFlow, Scikit-learn

Data Analysis Tools: Excel (Advanced Power Query, VLOOKUP, PivotTables), Tableau, Power BI (DAX), Looker, QlikView, Grafana

Databases: MySQL, PostgreSQL, MongoDB, SQL Server, Oracle, AWS Athena, Google Big Query, Redis

Cloud Technologies: AWS (S3, Redshift, Lambda, EC2), Azure (Synapse), Google Cloud (Big Query), Snowflake, Databricks

Big Data Technologies: Hadoop, Apache Spark, Apache Kafka, Apache Nifi, HDFS, Apache Oozie, Apache Hive, Sqoop

Data Science & Machine Learning: Regression, Classification, Clustering, Time Series Forecasting, ARIMA, LSTM, Predictive Modeling, Statistical Analysis

ETL & Data Engineering: Informatica, Alteryx, Apache Airflow, Mage, Control-M, ServiceNow, Talend, dbt, Data Modeling (ER Diagrams, Star/Snowflake Schema), Data Warehousing (OLAP, OLTP), Great Expectations

Automation & DevOps: Docker, Kubernetes, Jenkins, Git, CI/CD Pipelines, Unit Testing, SDLC, Apache Atlas, Flask API, Terraform, Bitbucket

Statistical & Mathematical Tools: SAS, SPSS, SRSS, Prometheus, Elastic Stac

Workflow & Collaboration Tools: JIRA, Agile, Scrum, Confluence, Trello

API Integration: REST APIs.

Cybersecurity & Governance: Data Governance, Cybersecurity Compliance (GDPR, CCPA), Real-time Monitoring, OCR, GIS, Query Optimization, Data Cleansing

Work Experience

Sr. Data Engineer | Wellington Management Company | Boston, MA

August 2022 – Present

Key Roles & Responsibilities:

- Enhanced **data quality and governance**, reducing financial reporting errors by 30% by implementing automated validation rules and anomaly detection.
- Implemented **AWS Lambda** functions and **EC2** instances to create a scalable architecture for financial data processing, enabling consistent performance during peak analysis periods
- Designed and maintained complex data models (**ER diagrams, star/snowflake schema**), improving investment data structuring and query performance.
- Leveraged **Azure Synapse Analytics and Azure Data Lake Storage** for scalable storage and parallel query execution to optimize financial data pipelines.
- Built and maintained data ingestion workflows using **Azure Data Factory and AWS Lambda**, ensuring efficient batch and real-time processing of financial datasets.
- Used **Google BigQuery** to analyze structured data from market feeds and transaction records, optimizing query cost and runtime.
- Integrated **RESTful APIs** with internal applications to automate data fetch from external financial services, enabling real-time dashboards.
- Engineered robust ETL workflows using **Informatica, dbt, and Airflow**, ensuring secure and consistent data movement between **data lakes and warehouses (Redshift, Snowflake)**.
- Automated **ETL** workflows, integrating **banking and investment** data from multiple sources, reducing manual effort and ensuring 99.9% data accuracy.

- Implemented robust **CI/CD** pipelines using **Jenkins and Git**, reducing deployment errors and enabling seamless integration of new code into production environments for data analytics applications.
- Expanded the use of **Docker** to containerize data processing applications, improving deployment consistency across environments and reducing configuration issues by 40%.
- Applied **TensorFlow** deep learning frameworks to develop **neural network** models for market prediction, achieving higher accuracy than traditional statistical methods.
- Developed robust **SQL** queries, reducing query execution time, enhancing financial data extraction, analysis, and reporting across SQL Server, Oracle, and MySQL.
- Utilized Azure Analysis Services to enable **OLAP** capabilities, enhancing interactive dashboards in Power BI and Looker.
- Integrated **Microsoft Fabric** to establish unified data observability and governance, improving cross-team visibility and data lineage.
- Used **PolyBase** in Azure Synapse Analytics for high-throughput data ingestion from external sources, reducing query processing time by 30%.
- Collaborated with **ServiceNow** to automate compliance alerting and incident tracking, supporting audit traceability and data governance.
- Designed and deployed scalable data lake architecture on Azure, integrating **Azure Data Lake Storage Gen2** and Databricks, enabling high-performance data processing for advanced analytics.
- Developed ETL pipelines using **Azure Data Factory**, executing data transformations and orchestrating ingestion workflows to centralize financial and investment data from MySQL and S3 into **Snowflake**.
- Built Spark-based pipelines using PySpark and Azure Databricks to conduct **Exploratory Data Analysis (EDA)** and transform massive financial datasets, reducing ETL job execution time by 40%.
- Ensured GDPR & CCPA compliance by implementing **Azure Purview** for data cataloging and **Azure Key Vault** for secure key management, enforcing governance and traceability.
- Integrated **Snap Logic** pipelines for syncing investment datasets into Salesforce dashboards, enabling seamless real-time reporting.
- Processed real-time data streams using **Apache Kafka** and Azure Stream Analytics for anomaly detection in high-frequency trade data, increasing compliance flag detection accuracy.
- Enhanced model scalability by deploying ML models within Azure Kubernetes Service (AKS), integrated with Azure Databricks **MLflow** for model tracking and experimentation.
- Created interactive visualizations with **Looker and QlikView** to provide alternative views of investment data, increasing user adoption of self-service analytics.
- Integrated **AWS services (S3, Redshift)** for scalable data storage and analytics, cutting down processing time for large financial datasets.
- Developed predictive models using **Python & scikit-learn**, improving risk assessment accuracy, supporting investment decision-making.
- Created an automated trade **anomaly detection** system using supervised machine learning algorithms, identifying potential **compliance** issues 75% faster than manual auditing processes.
- Deployed a **Python-based attribution analysis** framework that decomposed **portfolio performance** into factor contributions, enhancing investment strategy evaluation and client reporting.
- Engineered a custom **time-series forecasting** model using **ARIMA and LSTM** neural networks, achieving 18% higher accuracy in market volatility predictions compared to traditional methods.
- Led the design and deployment of a firmwide **data lake architecture on Azure**, consolidating disparate data sources and reducing data retrieval time while enabling advanced cross-asset analytics.
- Designed and deployed a comprehensive **data lineage** tracking system, ensuring full transparency in data transformations and supporting regulatory reporting requirements with 100% audit traceability.
- Designed and implemented a comprehensive bond analytics platform using **R and Shiny**, providing interactive yield **curve analysis** and scenario testing capabilities for fixed-income portfolios.
- Built an automated financial statement analysis tool using **OCR and machine learning**, extracting and standardizing financial data from diverse reporting formats, reducing manual processing time.
- Architected a real-time market sentiment analysis tool using **API** integrations with Twitter and financial news sources, providing traders with early market movement indicators and improving trade timing.
- Applied **SPSS and SRSS** for comprehensive statistical analysis of portfolio risk profiles, delivering insights that improved risk-adjusted returns by 12%.

Tech Stack: Python (Pandas, Scikit-learn, NumPy, Matplotlib), SQL, Power BI, Tableau, AWS (S3, Redshift), Azure, SQL Server, Oracle, MySQL, R (Shiny), Scikit-learn, TensorFlow (LSTM), Apache Spark, ARIMA, Docker,

Jenkins, Git, GDPR, CCPA, OCR, Data Modeling (ER Diagrams, Star/Snowflake Schema), Real-time Monitoring, GIS.

Data Quality Analyst |Void Main Technologies| Hyderabad, India

February 2022 – June 2022

Key Roles & Responsibilities:

- Orchestrated the creation of an air quality prediction model using Python, utilizing **Pandas, and Scikit-learn** for **cleaning and preprocessing** 100,000+ records which improved prediction accuracy compared to existing methods.
- Spearheaded the creation and upkeep of a centralized **metadata management** system, ensuring consistency across reporting and analytics.
- Conducted **exploratory data analysis** and created visualizations with **Matplotlib and Seaborn** to enable real-time pollutant monitoring and informed decision-making.
- Applied **advanced feature engineering** techniques, such as **outlier detection and normalization**, resulting in a 20% improvement in the model's predictive performance.
- Established effective project management practices using **JIRA and Confluence**, improving team collaboration and ensuring timely delivery of analytics projects.
- Leveraged **Apache Airflow** to orchestrate **ETL** workflows, automating the ingestion of air quality data from **APIs** and CSV files into a centralized **PostgreSQL** database, reducing pipeline latency.
- Built and orchestrated data pipelines in **Google Cloud Platform (BigQuery, Cloud Storage)** for high-volume environmental data processing.
- Ingested semi-structured environmental data using **Apache Sqoop** and processed it into Hive tables over HDFS for efficient querying.
- Scheduled batch ETL jobs using **Apache Oozie** to aggregate and transform daily pollutant metrics, reducing manual intervention.
- Processed large air quality datasets in **Avro and ORC** formats, optimizing storage footprint and improving processing speed by 20%.
- Configured **Snowflake data warehouse** and implemented **star/snowflake** schema designs for fast, scalable air quality analytics.
- Utilized **Apache Kafka and GCP Pub/Sub** to manage real-time ingestion from IoT devices, enabling sub-second latency for alerting systems.
- Developed API integrations using **Flask and FastAPI** to stream cleaned datasets to machine learning models and reporting tools.
- Designed and implemented ETL pipelines using dbt and Airflow, automating data transformations and reducing manual pipeline failures.
- Utilized **PySpark on Databricks** to process and clean large-scale air quality datasets (e.g., 100k+ records), enabling distributed computing for faster **data transformations and aggregations**.
- Implemented **data quality** rules using Great Expectations, validating pollutant metrics (e.g., PM2.5, NO2) for consistency and accuracy, achieving a 98% **compliance** rate across datasets.
- Designed and deployed a **Snowflake data warehouse** schema to store and query historical air quality and meteorological data, optimizing retrieval times for analytical workloads.
- Built **real-time** data monitoring pipelines with **Apache Kafka**, streaming live pollutant readings from **IoT** sensors into a **Redis cache** for immediate quality checks and anomaly detection.
- Processed large-scale air quality data using PySpark in Azure Databricks, applying advanced Spark tuning and memory optimization for faster aggregation and transformation.
- Implemented **Azure Data Factory** pipelines to orchestrate ingestion from external APIs into ADLS and PostgreSQL, supporting structured and unstructured datasets.
- Built real-time data ingestion and streaming pipelines using **Apache Kafka and Spark Structured Streaming** for continuous monitoring of IoT sensor data.
- Applied **Azure Synapse Analytics** for processing batch data and created Snowflake models using star/snowflake schema, enhancing retrieval speed and enabling cross-platform querying.
- Developed **ELT** pipelines in Snowflake using Azure Data Factory and **Delta Live Tables**, enabling historical data tracking and improving time-to-insight by 50%.
- Integrated **Apache Atlas** for data lineage documentation and governance compliance, ensuring end-to-end traceability.
- Developed SQL-based data profiling scripts in **Big Query** to assess completeness and uniqueness of air quality records, identifying and resolving 10% of previously undetected data gaps.

- Integrated **dbt (Data Build Tool)** to transform raw air quality data into curated analytical models, enabling reproducible data pipelines and improving team collaboration on analytics projects.
- Used **Docker** containers to deploy a **Flask-based API** for serving cleaned air quality datasets to downstream **machine learning** models, ensuring consistent data delivery across environments.
- Applied **Tableau** for advanced data visualization, creating interactive dashboards to track **AQI** trends and pollutant concentrations, enhancing stakeholder insights and **decision-making** processes.
- Conducted **data lineage** tracking with **Apache Atlas**, documenting the flow of air quality data from source to model input, ensuring traceability and compliance with data governance standards.

Tech Stack: Python (Pandas, Scikit-learn, Matplotlib, Seaborn), SQL, Tableau, Apache Airflow, PySpark, Databricks, Snowflake, Apache Kafka, Redis, Google Big Query, dbt, Docker, Flask API, Great Expectations, PostgreSQL, Apache Atlas, Real-time Monitoring

Data Analysis Consultant | Metafin Consulting | Hyderabad, India

January 2020 – January 2022

Key Roles & Responsibilities:

- Streamlined **tax filing** processes through programmed data checks, reducing client errors by 15% and optimizing SQL databases for improved data management of large tax datasets.
- Revamped tax reporting process using Excel automation (**Power Query, VLOOKUP**) and Python scripts, turning a 5-day manual review into a 1-day automated workflow while improving compliance accuracy.
- Crafted user-friendly **Power BI and Tableau** dashboards for visualizing **key tax metrics**, increasing client engagement and facilitating faster decision-making.
- Implemented **Kubernetes** to orchestrate containerized applications for **financial modeling**, enhancing scalability and reducing infrastructure costs by 25%.
- Collaborated with **cross-functional** teams to translate tax and compliance requirements into data-driven solutions, ensuring regulatory compliance.
- Designed SQL-based **ETL** pipelines to process large-scale financial datasets, improving data availability for consulting projects by 20%.
- Automated **data validation** for client deliverables using **Python (Pandas)**, reducing manual errors by 18% and ensuring high-quality outputs.
- Developed **Power BI** dashboards (**DAX**) to track **key performance indicators (KPI's)** for tax compliance, enhancing client presentations and decision-making.
- Utilized **Python (NumPy, Stats models)** to perform statistical analysis on historical tax data, identifying trends that saved clients an average of 10% on filings.
- Created **Excel VBA macros** to automate repetitive **data reconciliation** tasks, cutting processing time by 15 hours per week for the team.
- Created and deployed infrastructure using **Terraform**, automating provisioning for scalable data processing environments across AWS and Azure.
- Used **Bitbucket** for source control and collaboration on ETL pipelines, **dbt** models, and analytics codebases, improving version control.
- Built a **PostgreSQL** database schema to centralize client financial data, reducing query times by 25% for analytical reports.
- Automated report generation using **Python (Report Lab)** and scheduled workflows with Apache Airflow, delivering client reports 40% faster.
- Utilized **AWS S3 and Redshift** to store, transform, and analyze large-scale financial datasets for compliance and audit readiness.
- Developed secure data ingestion pipelines integrating **REST APIs**, Excel files, and SQL Server into centralized **PostgreSQL and Redshift** databases.
- Implemented ETL processes using **Talend and Airflow**, reducing batch job run times and improving audit trails.
- Migrated existing SQL workflows to **Azure Data Factory**, improving reliability, scalability, and monitoring of data jobs.
- Optimized SQL Server queries with **indexing and partitioning**, improving performance by 20% for high-volume tax datasets.
- Designed ETL workflows in **Azure Data Factory and Informatica** to process large-scale financial records, implementing automation and **dbt** for transformation logic and version control.
- Migrated and processed tax-related datasets from **Excel** and on-prem SQL Server into **Snowflake** using optimized **PySpark** jobs, improving load efficiency by 35%.
- Implemented distributed data processing with **Azure Databricks**, leveraging Spark SQL and PySpark for scalable transformation of structured and semi-structured tax records.
- Utilized **Airflow** to orchestrate batch and real-time workflows, integrating **Kafka**-based event pipelines with **REST APIs** to enable anomaly alerts and client notifications.
- Automated CI/CD deployment with **Jenkins**, using **Docker and Kubernetes** for container orchestration, reducing deployment time by 40%.

- Enabled secure and compliant data handling by integrating **Azure Key Vault** with authentication flows and enforcing fine-grained access control using **Active Directory (AD)**.
- Conducted performance tuning of SQL and Spark queries in **Synapse** and **Snowflake**, achieving 20% faster data extraction across compliance datasets.
- Developed machine learning models (**XGBoost**) in Python to predict tax audit risks, enabling proactive client recommendations.
- Integrated **AWS S3** with **Python** scripts to manage and store large datasets securely, supporting scalability for Metafin's growing client base.
- Used **Power Query in Excel** to streamline data imports for **ad-hoc analysis**, reducing preparation time by for urgent client requests.
- Conducted **data cleansing** on unstructured financial records with **Pandas**, improving data accuracy for downstream analytics by 15%.
- Built **real-time compliance** monitoring tools in Power BI, alerting clients to tax deadlines and reducing late submissions by 20%.
- Analyzed regulatory updates via **web scraping (Python, BeautifulSoup)**, ensuring datasets reflected current tax laws for client reports.
- Implemented **anomaly detection** scripts in Python (SciPy) to flag irregularities in financial data, cutting error rates by 12%.
- Used **Git** for **version control** of analytical scripts and SQL queries, enhancing team collaboration and code reliability.
- Designed interactive Excel dashboards with **PivotTables** for small business clients, simplifying financial tracking without advanced tools.
- Developed predictive models in **Python (TensorFlow)** to estimate tax liabilities, improving forecasting accuracy by 12% for clients.
- Supported process optimization projects by analyzing workflow data in **R**, identifying bottlenecks that saved 8 hours weekly.

Tech Stack: Python (Pandas, NumPy, Stats models, Report Lab, BeautifulSoup, SciPy, TensorFlow, XGBoost), SQL, Power BI (DAX), Tableau, Excel (Power Query, VLOOKUP, PivotTables, VBA), Apache Airflow, PostgreSQL, SQL Server, AWS S3, R, Git, REST APIs, Real-time Monitoring.

Jr.BI Analyst| Universal Financials | Hyderabad, India

January 2016 – December 2019

Key Roles & Responsibilities:

- Assisted in developing financial dashboards using **Power BI**, improving reporting efficiency and enhancing data-driven decision-making.
- Supported the automation of **financial reporting** processes using SQL and Python, reducing manual errors by 40% and saving valuable time.
- Conducted **variance analysis** on financial reports, identifying discrepancies that led to improved accuracy in financial planning.
- Collaborated with senior analysts to integrate financial data from multiple sources into a centralized reporting system, improving data accessibility.
- Performed **ad-hoc financial data analysis** to uncover inefficiencies, supporting business decisions that optimized resource allocation.
- Automate routine data reporting tasks and processes using **SQL** queries, **Python**, or **PowerShell** scripts to increase efficiency.
- Automated financial data refresh from external **APIs** into **Azure Blob Storage** and **AWS S3**, supporting real-time dashboarding.
- Developed SQL queries for ingestion into **Google Big Query**, reducing data lag and storage cost for recurring reports.
- Processed **Sequence files** into Azure Data Lake and Snowflake, enabling downstream analytics with significantly lower data latency.
- Implemented **Change Data Capture (CDC)** mechanisms to track transactional changes and keep Snowflake data warehouse synchronized in near real time.
- Built end-to-end data pipelines using ETL tools and custom Python scripts, consolidating Excel, CSV, and API inputs into a single reporting data warehouse.
- Integrated **Azure Data Lake Gen2** with **Power BI** to enable **ad-hoc analytics** across structured and semi-structured financial data.
- Developed SQL queries to **extract and aggregate** financial data from large datasets, reducing report generation time by 25%.
- Automated repetitive data entry tasks using **Python (Pandas)**, cutting manual processing time by 15 hours per week for the team.

- Created Excel-based financial models (**VLOOKUP, PivotTables**) to track expense trends, improving budget monitoring for small business clients.
- Assisted in building Power BI reports (**DAX**) to visualize tax refund statuses, increasing client engagement by 20%.
- Supported data cleansing efforts using **Python (NumPy)** to ensure accuracy in financial datasets, reducing errors by 10%.
- Collaborated with the IT team to implement a My-SQL database for storing client financial records, enhancing data retrieval speed by 30%.
- Used Tableau to create basic visualizations of revenue streams, aiding senior analysts in presenting insights to management.
- Performed trend analysis on historical financial data with **Python (Matplotlib)**, identifying patterns that informed quarterly forecasts.
- Assisted in automating monthly financial summaries using Python scripts and **Excel macros (VBA)**, saving 12 hours of manual effort monthly.
- Conducted **data validation** checks on tax-related datasets using SQL, ensuring compliance with regulatory standards for client reports.
- Supported the integration of external data sources (e.g., bank statements) into Power BI dashboards via **Power Query**, streamlining reporting workflows.
- Helped develop a basic revenue prediction model in **Python (Scikit-learn)**, improving forecast accuracy by 8% for internal planning.
- Built foundational ETL workflows in **Azure Data Factory and SSIS**, migrating legacy financial data into Snowflake for centralized reporting.
- Developed **PySpark** scripts to automate data transformations, achieving 25% reduction in report generation time across financial departments.
- Assisted in building streaming pipelines with **Kafka and Redis** for real-time financial alert systems, improving time-to-response for anomalies.
- Implemented star schema models and optimized SQL queries in **Snowflake** and PostgreSQL to support OLAP-style dashboards in Power BI.
- Created **ad-hoc SQL** reports to support client tax filing inquiries, reducing response time by 20% for the business intelligence team.

Tech Stack: Python (Pandas, NumPy, Matplotlib, Scikit-learn), SQL, Power BI (DAX), Tableau, Excel (Power Query, VLOOKUP, PivotTables, VBA), MySQL, Git.

EDUCATION

The University of Texas at Dallas

Richardson, TX